# Artificial Intelligence (AI)

# LAB# 1

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**CMS: 023-23-0115**

## Task 1:

### Facts:

```
likes (mark, hunger).
likes (mark, cars).
likes (mark, people_who_like_him).
tall(dany).
short(napoleon).
sour(lemon).
sister (susan, john).
plays (charlie_parker, piano).
musical_instruments(piano).
play_with_friend (shaquille, basketball).
smart(sam).
smart(rida).
```

### Queries:

```
likes (mark, hunger).
likes (mark, cars).
likes (mark, cats).
tall (dany).
tall (napoleon).
short (dany).
short (napoleon).
sour (apple).
sour (lemon).
sister (susan, john).
sister (susan, mark).
plays (charlie_parker, piano).
plays (charlie_parker, X).
musical_instruments(piano).
musical_instruments(X).
play_with_friend (shaquille, basketball).
play_with_friend (shaquille, football).
smart (sam).
smart (rida).
```

SWISH File Edit Examples Help

Program

```
radhika_kumari.
023-23-0115.

likes(mark, hunger).
likes(mark, cars).
likes(mark, people_who_like_him).

tall(dany).
short(napoleon).
sour(lemon).
sister(susan, john).
plays(charlie_parker, piano).
musical_instruments(piano).
play_with_friend(shaquille, basketball).
smart(sam).
```

```
likes(mark, hunger).
true
likes(mark, cars).
true
likes(mark, cats).
false
tall(dany).
true
tall(napoleon).
false
short(napoleon).
true
sour(apple).
false
```

SWISH File Edit Examples Help

Program

```
radhika_kumari.
023-23-0115.

likes(mark, hunger).
likes(mark, cars).
likes(mark, people_who_like_him).

tall(dany).
short(napoleon).
sour(lemon).
sister(susan, john).
plays(charlie_parker, piano).
musical_instruments(piano).
play_with_friend(shaquille, basketball).
smart(sam).
```

```
sour(lemon).
true
sister(susan, john).
true
sister(susan, mark).
false
plays(charlie_parker, piano).
true
plays(charlie_parker, X).
X = piano
musical_instruments(piano).
true
musical_instruments(X).
X = piano
```

SWISH File Edit Examples Help

Program

```
radhika_kumari.
023-23-0115.

likes(mark, hunger).
likes(mark, cars).
likes(mark, people_who_like_him).

tall(dany).
short(napoleon).
sour(lemon).
sister(susan, john).
plays(charlie_parker, piano).
musical_instruments(piano).
play_with_friend(shaquille, basketball).
smart(sam).
smart(rida).
```

```
X = piano
musical_instruments(piano).
true
musical_instruments(X).
X = piano
play_with_friend(shaquille, basketball).
true
play_with_friend(shaquille, football).
false
smart(sam).
true
smart(rida).
false
smart(rida).
true
```

## Task 2:

### Facts:

```
has(jack, apples).

has(ann, plums).

has(dan, money).

fruit(apples).

fruit(plums).
```

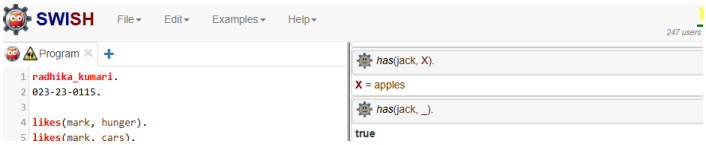
### Queries

- What has Jack?

  ```
  has (jack, X).
  ```

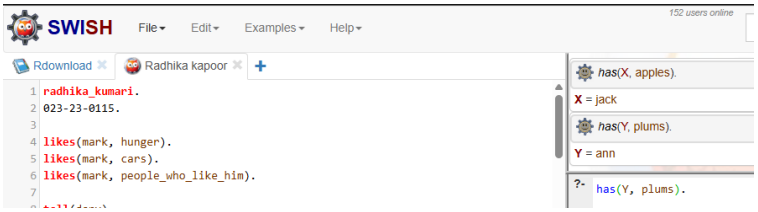
- Does jack have something?

  ```
  has (jack, _).
  ```



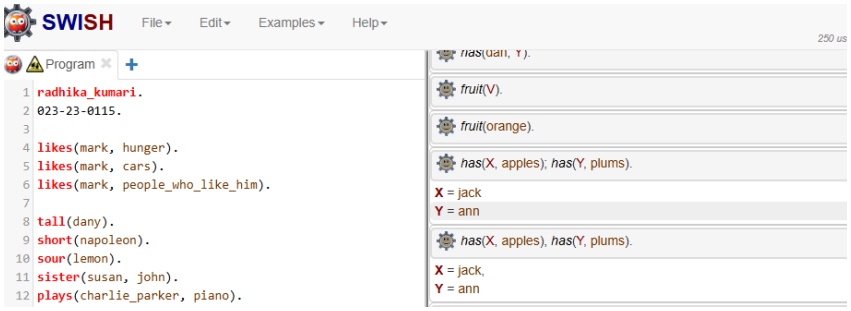

- Who has apple and who has plum?

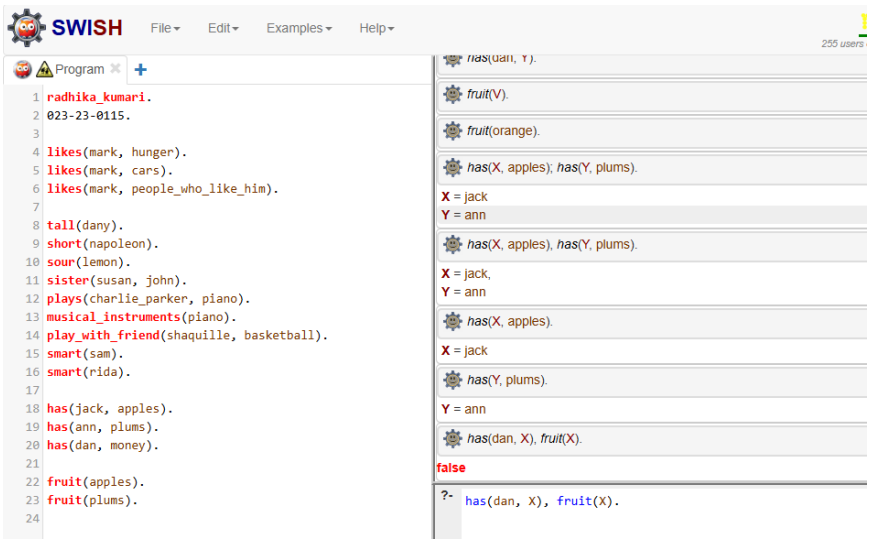
  ```
  has (X, apples).

  has (Y, plums).
  ```



- <u>Does someone have apple and plum?</u>

```
has (X, apples); has (Y, plums).
```




- <u>Has Dan fruits?</u>

```
has (dan. X), fruit(X).
```